

No. of Functional Features Included in the Solution

Date	05 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Dashboard Page	Renders comparative performance metrics and inline EDA charts for all trained models.	<code>templates/home.html</code>	Done	15%
2	3-Step Predict Wizard	An intuitive form separating User Profiles, Personal Backgrounds, and Financial Details.	<code>templates/predict.html</code>	Done	20%
3	Input Form Validation	Ensures all required fields are correctly completed before submitting data to the server.	Frontend JS Script	Done	10%
4	Data Scaling	Applies trained <code>StandardScaler</code> parameters to incoming numeric arrays on-the-fly.	<code>app.py / scale1.pkl</code>	Done	15%
5	XGBoost Classification	Runs prediction inference on standardized vectors using a tuned XGBoost Classifier.	<code>app.py / rdf.pkl</code>	Done	20%
6	SQLite Transaction DB	Commits applicant profiles, loans, credit history, and	<code>models.py</code>	Done	10%

		prediction results via SQLAlchemy.			
7	History Log Tables	Displays a chronological log of all past predictions with detail parameters and statuses.	templates/history.html	Done	5%
8	Record Deletion	Allows administrators to purge individual transaction logs via cascade database deletion.	app.py Route	Done	5%

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home Dashboard layout, 3-Step Predict Wizard, Results Display template, History Log list table.
2	Backend / Logic	3	On-the-fly parameter encoding, Features Scaling logic, XGBoost Inference executor.
3	Database / Storage	1	SQLite relational database models (user, ApplicantProfile, LoanApplication, PredictionResult).

4	API / Integration	1	Model Pickle binary file mapping helper (load_prediction_model).
5	Security / Authentication	1	Profile role classification check (Credit Officer / Analyst / Applicant) on form validations.